

VIII. Mini Project - Metro Interstate Traffic Analytics

AIM

To clean, process, analyze, and visualize Metro Interstate traffic-volume data using Python, and build analytical dashboards using Python, Power BI, and Tableau to identify traffic patterns and relationships between traffic volume, time, weather, temperature, and holidays.

DOMAIN

Transportation and Traffic Analytics

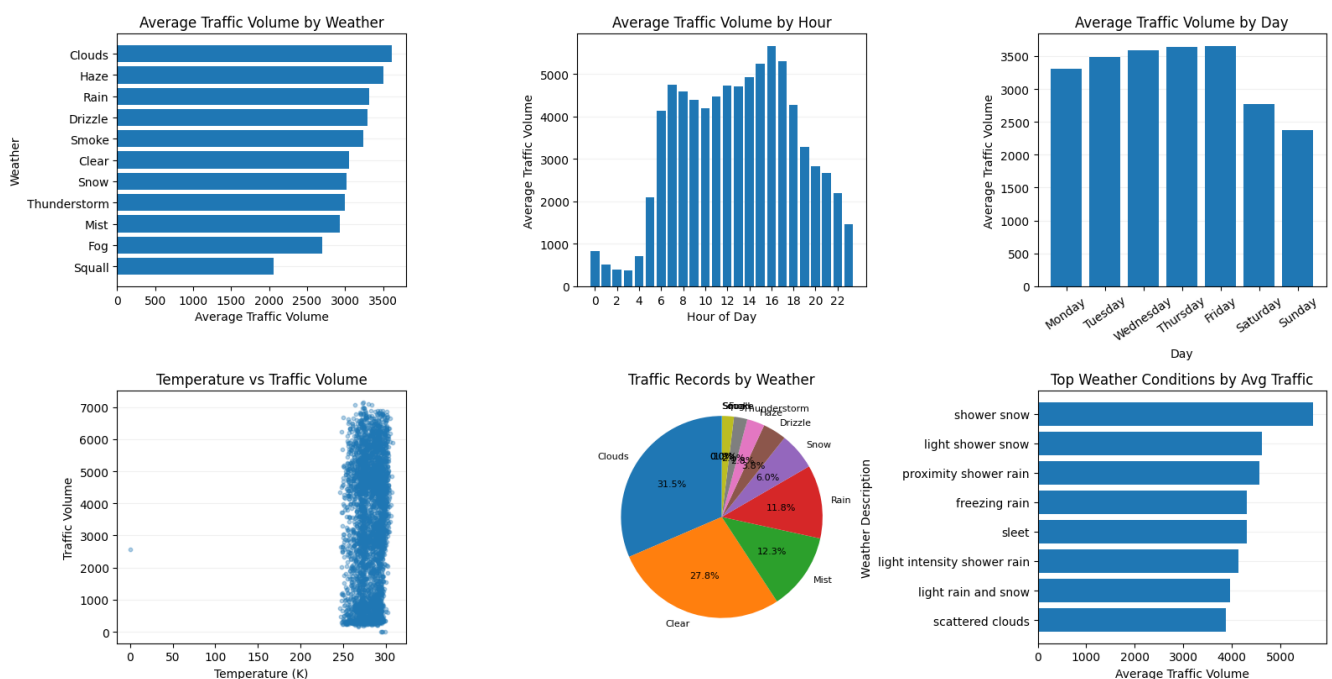
DATASET

The Metro Interstate Traffic Volume dataset contains 48,204 traffic observations and 9 variables. The data covers 02 Oct 2012 to 30 Sep 2018. The main variables include holiday, temperature, rain, snow, cloud coverage, weather condition, weather description, date/time, and traffic volume.

PROBLEM STATEMENT

Traffic volume changes significantly with time, weather conditions, and other environmental factors. A data-driven visualization system is required to analyze hourly and daily traffic patterns, compare traffic under different weather conditions, examine temperature versus traffic volume, and present the findings through clear and interactive

Metro Interstate Traffic Analytics



PowerBI:

Metro Interstate Traffic Analytics

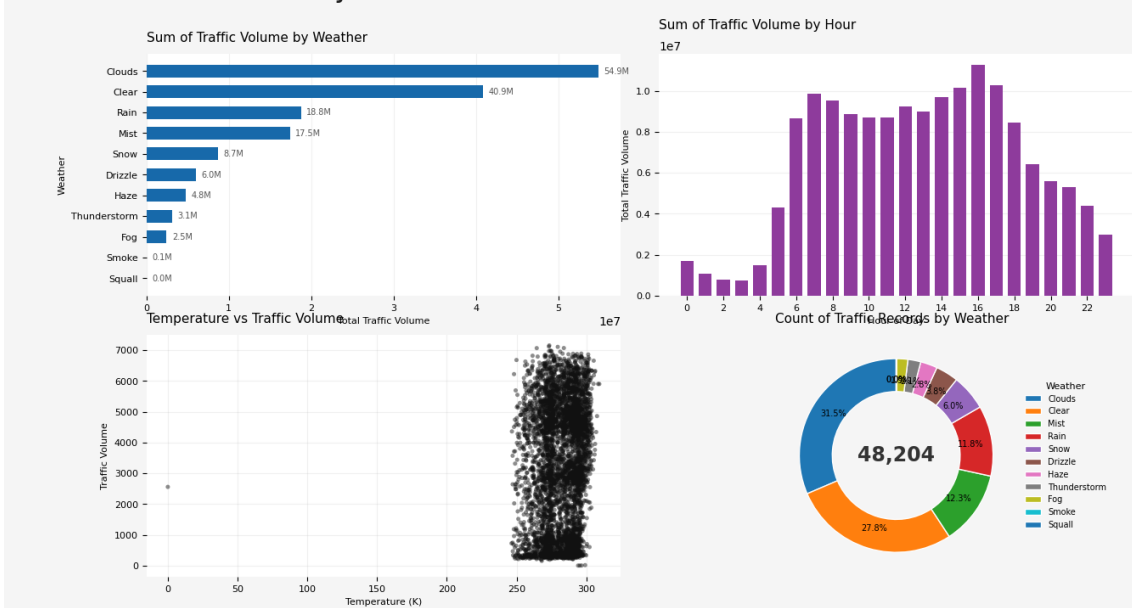
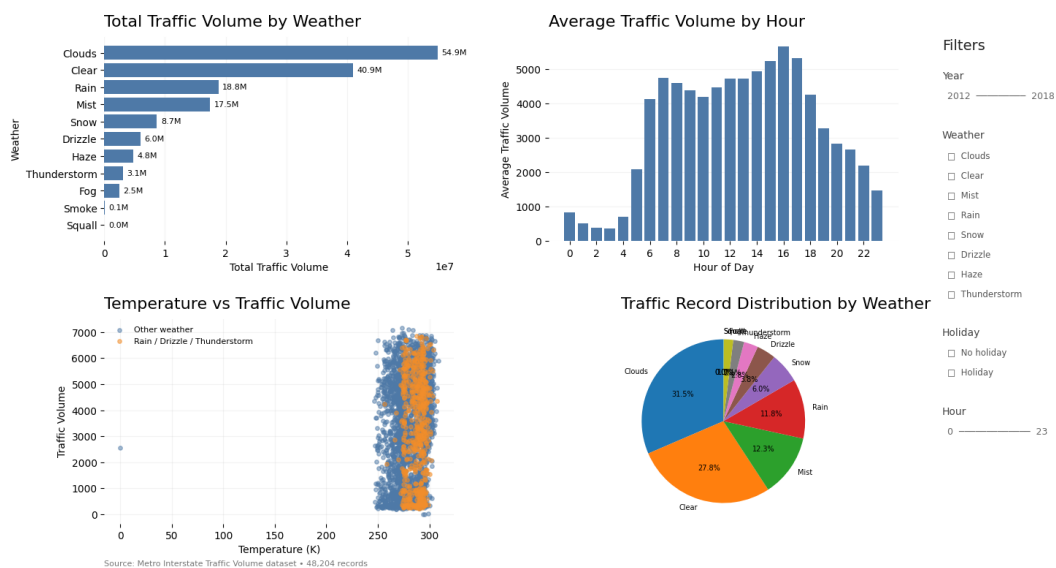


Tableau:

Metro Interstate Traffic Analytics Dashboard



RESULT

Thus, the Metro Interstate Traffic Analytics mini project was designed and implemented successfully using Python, Power BI, and Tableau. The visualizations provide analysis of traffic volume by weather and hour, daily traffic patterns, temperature versus traffic volume, and the distribution of traffic records across weather conditions.